

Exercise 1:

Assuming tables:

```
1  CREATE TABLE Customers (  
2      CustID NUMBER PRIMARY KEY,  
3      Name VARCHAR2(100),  
4      Age NUMBER,  
5      Balance NUMBER(10,2),  
6      IsVIP VARCHAR2(5)  
7  );  
8  
9  CREATE TABLE Loans (  
10     LoanID NUMBER PRIMARY KEY,  
11     CustID NUMBER,  
12     InterestRate NUMBER(5,2),  
13     DueDate DATE,  
14     FOREIGN KEY (CustID) REFERENCES Customers(CustID)  
15 );  
16  
17 INSERT INTO Customers VALUES (1, 'John', 65, 15000, 'FALSE');  
18 INSERT INTO Customers VALUES (2, 'Alice', 45, 8500, 'FALSE');  
19 INSERT INTO Customers VALUES (3, 'Bob', 72, 22000, 'FALSE');  
20 INSERT INTO Customers VALUES (4, 'Emma', 30, 12000, 'FALSE');  
21 INSERT INTO Customers VALUES (5, 'David', 61, 9500, 'FALSE');  
22  
23 COMMIT;  
24  
25 INSERT INTO Loans VALUES (101, 1, 9.5, SYSDATE + 10);  
26 INSERT INTO Loans VALUES (102, 2, 10.2, SYSDATE + 45);  
27 INSERT INTO Loans VALUES (103, 3, 8.8, SYSDATE + 20);  
28 INSERT INTO Loans VALUES (104, 4, 9.1, SYSDATE + 5);  
29 INSERT INTO Loans VALUES (105, 5, 10.0, SYSDATE + 28);  
30  
31 COMMIT;
```

Scenario 1:

```
35 BEGIN
36   FOR cust IN (SELECT CustID, age FROM Customers) LOOP
37     IF cust.age > 60 THEN
38       UPDATE Loans SET InterestRate = InterestRate-1
39       WHERE custID = cust.custID;
40     END IF;
41   END LOOP;
42
43   DBMS_OUTPUT.PUT_LINE('Loan rates updated');
44 END;
45 /
```

Output:

Loan rates updated

Scenario 2:

```
50 BEGIN
51   FOR cust IN (SELECT CustID, Balance FROM Customers) LOOP
52     IF cust.Balance > 10000 THEN
53       UPDATE Customers
54       SET IsVIP = 'TRUE'
55       WHERE CustID = cust.CustID;
56     END IF;
57   END LOOP;
58
59   COMMIT;
60
61   DBMS_OUTPUT.PUT_LINE('VIP status updated');
62 END;
63 /
```

Output:

VIP status updated

Scenario 3:

```
67 BEGIN
68     FOR loan_rec IN (
69         SELECT c.Name, l.LoanID, l.DueDate
70         FROM Customers c
71         JOIN Loans l
72         ON c.CustID = l.CustID
73         WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30
74     ) LOOP
75
76         DBMS_OUTPUT.PUT_LINE(
77             'Reminder: ' || loan_rec.Name || ', your loan with Loan ID: ' ||
78             loan_rec.LoanID || ' is due on ' || TO_CHAR(loan_rec.DueDate,
79             'DD-MON-YYYY')
80         );
81
82     END LOOP;
83 END;
84 /
```

Output:

```
Reminder: John, your loan with Loan ID: 101 is due on 10-JUL-2026
Reminder: Bob, your loan with Loan ID: 103 is due on 20-JUL-2026
Reminder: Emma, your loan with Loan ID: 104 is due on 05-JUL-2026
Reminder: David, your loan with Loan ID: 105 is due on 28-JUL-2026
```